Investor challenge. Evidence-led version 2.

A candid decision brief and source-grounded delivery plan

5 decision tracks | 14 findings | 93 backlog IDs retained

Investor question	Current evidence	Unresolved decision
Inspectable engineering?	Source mechanisms and bounded historical demonstrations.	Current installed behavior and the selected integrated workflow.
Scientific advantage?	Complete retrospective PLGA matrix; all ten planned intervals include zero.	New-panel benefit/transfer; no established Pareto-specific superiority.
Product integration?	PLGA is a separate adapter; no LLM calls.	No end-to-end PLGA campaign-engine qualification supplied.
Commercial value?	Workflow composition/reuse is a hypothesis.	Buyer pain, current alternative, adoption, pricing and matched value evidence.
Defensible assets?	Versioned source and data provenance.	Chain of rights, accountable owners/budgets and incremental advantage.

Decision: deeper technical diligence and a bounded next proof are supportable. None of the five investment tracks is closed. Preparation task counts are not scientific or commercial derisking.

Two reproduced V1 defects are addressed in this revision: mutable hashes could bless a fake historical checker, and a README could qualify an owner-approval task. The original PLGA paper and results remain unchanged.

18 October 2026 | Critical technical investor review | Draft for review

Amphora has a credible engineering proposition: make bounded research workflows executable, keep model proposals separate from execution authority, and retain the inputs, decisions and outputs needed to inspect what happened. The repository offers concrete mechanisms and historical demonstrations to examine. The investment question is whether those mechanisms can support a useful real workflow, an independently valuable scientific outcome and a defensible business. Those are separate evidence obligations.

Version 2 is more demanding than version 1. It documents two reproduced verifier/status defects, fixes feasible tooling weaknesses, removes task-count readiness percentages and leads with five decision tracks. It does not manufacture better predictive performance, current production qualification, a buyer, exclusive rights or a commercial moat.

1. One-page current-state decision brief

Question	What is available now	What remains unestablished
Is there inspectable engineering substance?	Typed compilation, owner acceptance, retained context, search/reuse and replacement boundaries; scoped historical proof records	Current installed-host health, whole-stack Linux qualification and unrestricted autonomy
Is there real scientific computation?	Separate retrospective PLGA adapter, frozen protocol, complete comparisons, recorded replay and model-free checks	An end-to-end PLGA run through the deployed campaign engine
Is there a scientifically superior predictor?	Controlled exposed-corpus characterization of six arms and two learners	Pareto-specific superiority, equivalence or new-panel/ocular transfer; all ten planned intervals include zero
Is there measured customer value?	A technical hypothesis about composition, inspection and qualified reuse	Verified buyer pain, adoption, willingness to pay, pricing, matched workflow savings or retention
Is the packet investment-ready?	Detailed preparation, source pins and corrected integrity tooling	Approved investor prose, accountable role assignments, rights/disclosure decisions and a qualified external release

Decision: the work supports deeper technical diligence and a bounded next proof, rather than a conclusion that scientific or commercial risk has been retired. None of the five investment decision tracks is closed. A script pass is a statement about the named mechanical check, not a financing recommendation.

The 93 original task IDs remain traceable in the backlog. They are a work inventory, not a denominator for investment readiness. Uniform five-to-thirty-minute task estimates and the six-hour aspiration have been removed as uncalibrated scheduling inputs. Future work remains unestimated until a named person accepts a bounded scope and budget. This revision authorizes and delivers the review, feasible local fixes and a public draft PR; it does not execute the remaining backlog.

2. The consequential findings

Fourteen findings are recorded with severity, source revision/path/hash, investor impact, concrete fix, verification and disposition. The review distinguishes confirmed code defects, presentation weaknesses, missing evidence and unresolved ownership/commercial questions. Missing evidence is unverified; it is not declared false.

Confirmed defect RF-01: version 1 accepted a fake historical checker after its mutable mapping and manifest hashes were recomputed. The unchanged V1 verifier then returned `verified` with fabricated gate names. This demonstrated self-consistency, not a binding to the original checker. Version 2 anchors the approved policy/configuration to reviewed code, checks the exact payload allowlist before execution and verifies the original gate set, profile, limitations and result semantics. The coordinated-substitution regression must reject before running the substituted checker.

Confirmed defect RF-02: version 1 accepted future owner-approval task DD-010 as verified when only the root README was supplied as evidence. File existence was insufficient. Version 2 binds evidence to its bytes, supported task IDs, actual result/exit fields and current verifier code for mechanical checks. Preparer records cannot establish the protected role-approval tasks. These checks still rely on an accurately produced receipt; they do not identify a person cryptographically or prove that a fraudulent receipt producer is honest.

Investment evidence gaps: historical owner demonstrations and the isolated scientific benchmark do not establish integrated scientific product behavior. The internal PLGA result does not establish predictive lift or transfer. The packet supplies no verified buyer/value or exclusive-asset evidence. Version 2 makes those uncertainties first-class decision tracks with unfavorable-result branches rather than burying them behind a large task register.

Review provenance: the local preparer inspected the actual V1 PDF, ledger and code and executed the defect probes. An additional reviewer contributed reconciliation-based questions after Library artifact access was denied. That contribution is not represented as an independent full-artifact or code audit. This is not independent scientific validation, a legal opinion or a patentability review.

3. Five decision tracks with real stopping rules

DG-1 - Diligence inspectability

The adviser must be able to trace every principal statement to an exact source, scope and qualifier, and run the advertised reviewer path from a clean export. Require complete twelve-cell scientific results, all ten contrasts, every requested check and every final PDF page. Scope labels distinguish fresh integrity/arithmetic, historical process execution and omitted current integration. A failed required gate blocks that claimed release profile; a smaller profile must say precisely what it verifies.

Current state is partial: useful source and corrected tooling exist, but the investor brief/systems note and final role-reviewed release are not completed by this plan. If the evidence cannot be exported within rights/disclosure limits, publish a smaller supported route or remain draft. Do not weaken the original assertions to obtain a pass.

DG-2 - Product end-to-end integration

Choose one real scientific workflow and freeze its versions, inputs, data rights, public tool contracts and expected retained outputs. Qualify admission, tool execution, protective retention, accepted producer chain, output reconciliation and recovery through the actual installed owners. Require every named assertion, with zero unmapped input/output identities. Compare outputs to the separately frozen procedure; execution success alone does not establish scientific quality.

Current state is unverified for PLGA integration. The numerical examples remain valuable bounded software evidence. If integration fails, preserve the failed boundary and keep systems and scientific claims separate. POP-01 -> RG-01 -> ML-01 prerequisites remain owned by their existing plans; this review does not silently activate them.

DG-3 - Workflow value against the current alternative

First identify the buyer, actual user, bounded workflow and observed current alternative. Predeclare matched task definitions, primary effort/time measure, correctness threshold, onboarding/integration cost, sample/replication plan and decision margin. Those design values are currently undecided, not invented to make the packet appear complete. They must be chosen before outcomes are examined.

Retain all matched observations, failures and uncertainty. A faster different task is not a saving. If the value margin is not met, change or stop the use-case hypothesis. Do not convert bounded CPU time, a synthetic search comparison or packet-generation speed into assay savings, customer ROI or willingness to pay.

DG-4 - Independent future scientific outcome

Freeze one method/representation and comparator, a fixed information schedule and horizon, a predeclared estimand and margin, missing-data rules and assay-replicate uncertainty. Select genuinely unexamined, overlap-screened new-study PLGA data before seeing outcomes. Justify sample size and replication from the scientific question and variability; a blank or unsupported design blocks launch.

The current study remains post-pilot internal validation on the development-exposed Bao corpus. Report every eligible profile and endpoint under the frozen rule. An unfavorable result is retained. Do not search new seeds/grids on the exposed cohort to turn a documentation milestone into an apparent external win. Ocular or route-specific claims require their own applicable evidence.

DG-5 - Defensibility, provenance and accountable release

Map every selected contribution to its third-party foundations, Amphora-specific integration, evidence and rights disposition. Public source visibility and CC BY data are not evidence of an exclusive asset. Prior-art acknowledgement is not a patent or freedom-to-operate assessment. Ownership and reuse permissions need their actual records and responsible review.

Every activated future track needs an assigned person, approved budget and inclusion decision. Those remain unassigned/unapproved in this assessment. If rights or exclusivity cannot be supported, narrow the wording or payload rather than imply that this review resolved the question. The user's explicit instruction authorizes pushing this revised review branch; it does not turn an investor evidence release or scientific study into an approved completed gate.

4. Exercise matrix: what was actually run

Evidence object	Execution and scope	Permitted inference
PLGA matrix and replay	Separate retrospective adapter; no LLM calls; historical 780-fit matrix plus full replay	Inspectable representation evaluation on that corpus, not integrated platform qualification or independent new data
Historical campaign/curation/replacement records	Named synthetic/numerical workloads and historical owner/image versions	Those bounded recorded behaviors, not current host readiness or autonomous scientific quality
Current V2 reviewer checks	Fresh source/capsule integrity, original historical assertions and optional arithmetic, with exact receipts	Consistency with reviewed pins and requested calculations; no fresh service/process recovery
Future integrated scientific workflow	No new such run performed here	DG-2 proof must be produced separately
Future buyer/new-panel studies	No such measurements supplied or run	DG-3/DG-4 remain open

The generated scientific card retains all twelve stage-2 point estimates and all ten planned contrasts, with the development-exposed/no-superiority interpretation adjacent to both tables. Its 320 formulations, 4,870 retained observations and 3,910 suffix targets are different denominators. The 1,560 recorded fits are computational work, not 1,560 experiments or discoveries. Original scientific paper, source and results remain byte-identical.

5. A defensible contribution account

AiiDA's primary documentation already describes data/process provenance, workflow links and the distinction between data and logical provenance. LangGraph's primary documentation describes stateful agent orchestration, persistence and mixing deterministic and model-driven steps. Graphs, provenance and agent workflows are therefore foundations to credit, not a universal novelty claim. These two pages were checked on 18 October 2026; this is not an exhaustive prior-art search.

Amphora's specific claim is the inspected combination of typed research composition, owner-controlled acceptance, retained scientific context and bounded replacement/recovery examples. Whether this combination creates material incremental value or an exclusive/compounding asset is an investment hypothesis to test. The contribution/provenance matrix records what is implemented, what is sourced and what remains unverified; it does not infer competitor absence.

The commercial hypothesis ledger names a hypothetical buyer/user/workflow and possible measures while keeping interviews, pricing, ROI, adoption and retention unknown. Public Bao data are not described as proprietary assets. Existing source/data terms remain in force; this revision adds no blanket license or new ownership assertion.

6. Version 2 verification and trust boundary

The default reviewer route is offline after acquisition, starts no services or models and uses Python 3.12 standard-library dependencies. Optional bootstrap requires the separately supplied NumPy environment. Missing requested NumPy returns exit 3; evidence rejection returns 2; success means all requested required checks completed. Source and capsule reports enumerate what executed, which execution remains historical and what was omitted.

Trust begins with reviewed verifier/policy code and a trusted interpreter/local directory acquired from the identified Git revision. Fixed policy hashes bind the original map, source lock, allowlist and candidate registry. The capsule cannot introduce an extra executable, replace the original checker and rewrite its checksums, or change a declared transform while retaining the reviewed policy. An optional expected-manifest digest obtained outside the capsule adds an independently recorded integrity binding.

If the verifier and all its trust anchors are replaced, its own output is not an authentication mechanism. SHA-256 does not establish author identity, scientific truth or rights. Offline execution is not a safe sandbox. No signing credentials are created and no cryptographic publisher-authenticity claim is made.

Child checkers use isolated Python, bounded stdout/stderr, whole-report JSON validation and a shared timeout budget. The source profile runs the required capsule pass once. Canonical array pointers, phase membership, source ranges, task/evidence support and safe output resolution are checked. The supported local-directory model still does not claim a complete hostile-filesystem/TOCTOU defense.

The current PDF freshness check binds final source inputs, renderer and output bytes. A source/template/brand change invalidates the current render record. Visual acceptance is separately tied to the final PDF hash. This is a concrete current-document check; a universal future investor-prose assembler, signatures and complete production sandboxing are not claimed.

7. Critical path and concrete next deliverables

Keep one release owner and a small outcome path: **source/claim boundary -> reviewed proof cards and candid scope -> concise investor prose -> supported clean export -> final document/rights checks -> explicit distribution decision**. Current code hardening and review publication can complete now without executing DG-2/3/4 studies.

The smallest later investor artifact set is a two-page brief, a concise systems note, the unchanged eight-page PLGA paper, complete scientific card, evidence index and supported

reviewer command. A six-page systems version is an editorial target, not a reason to shrink type. Recording, FAQ, extra plots, live calls and new model replay are optional after the core qualifies. No guessed date or uniform task-minute budget is used as acceptance evidence.

All 93 task IDs remain available for implementation traceability. Their states distinguish inspected preparation, fresh mechanical checks and future approval/research. Role placeholders are not appointments. Protected approval tasks cannot be completed by an arbitrary existing file or a preparer receipt. No task-count percentage is presented as investor readiness.

8. Release and publication contract

This version is published on a branch with a draft PR for review. It does not merge itself or send the repository to investors. Preserve concurrent `main` changes; verify exact remote SHA and anonymous file access after pushing. The public payload includes the revised plan, candid findings/dispositions, updated tooling/tests, generated review PDF and sanitized source provenance. It excludes raw attachments, private email/Drive links, credentials, raw datasets, third-party full texts and unrelated historical operational dumps.

Original BP-09 pins and assertions remain unchanged. Original PLGA paper/results remain unchanged. Selected old receipt bytes may be checked locally in the internal capsule, but disclosure/rights for a later curated investor capsule are separate. Documentation fixes resolve representation and tooling defects; only new appropriate evidence can resolve integration, scientific, commercial or ownership questions.

9. Reviewable evidence and remaining decisions

The findings JSON and generated red-team report retain V1 source identities and the original mutation outcomes. V2 receipts list actual tests, code scope, interpreter/platform, requested arithmetic and final PDF QA. The changelog states which defects are fixed and which investor questions are only made explicit. Current-state and five-track assessments remain machine readable.

The remaining decisions are concrete: select the investor audience/claims and rights-reviewed payload; assign accountable people and budget; choose the one integrated workflow; define the buyer/current-alternative pilot; and freeze the independent scientific question and margin. The draft PR invites review of this stronger evidence plan, without implying those decisions or outcomes already exist.

Candid findings and disposition

RF-01 | HIGH | Mutable hashes could launder a substituted historical checker

Impact: A fake twelve-gate checker was accepted after its mutable mapping/manifest hashes were recomputed. This invalidates any claim that V1 necessarily ran the original assertions under such tampering.

Fix / disposition: Fixed reviewed policy hashes and exact payload allowlist are checked before execution; full original gate/profile/limitation semantics checked; use reviewed Git verifier and optional externally obtained manifest digest. [resolved_in_scoped_local_tests]

Verification / remaining proof: A coordinated fake-checker substitution with mutable hashes recomputed must reject before execution.

Source: e1198952 / capsule.py / lines 47-85

RF-02 | HIGH | A README could qualify an owner approval task

Impact: V1 accepted DD-010 as verified when its only evidence was the root README; evidence existence was mistaken for acceptance.

Fix / disposition: Evidence registry binds bytes, supported task IDs, result/exit status and current verifier code; protected role-approval tasks cannot be auto-promoted by preparer records. [resolved_in_scoped_local_tests]

Verification / remaining proof: README-only promotion, failed receipt and stale code binding reject.

Source: e1198952 / core.py / lines 197-248

RF-03 | HIGH | 93 tasks and preparation percentages obscure the investment decision

Impact: The 27-page packet showed 27.96% implementation and 26.88% verification for a mixed future register; these quantities do not measure scientific, product or commercial derisking.

Fix / disposition: Retain the 93 task IDs as a backlog; lead V2 with one current-state page and five decision tracks. Remove task-count readiness percentages and unjustified uniform effort ranges. [implemented]

Verification / remaining proof: Current-state and decision validators report zero investment gates met, with separate technical receipts.

Source: e1198952 / PROGRESS.md / lines 1-19

RF-04 | HIGH | The benchmark has not qualified the deployed product

Impact: The PLGA study is an isolated adapter, while owner/curation/recovery examples are different synthetic/historical scenarios.

Fix / disposition: Explicit exercise matrix and DG-2 frozen integration gate; no claim of an end-to-end PLGA engine run. [disclosed_future_proof]

Verification / remaining proof: Every scientific proof card says isolated adapter/no LLM; an integration claim requires its own actual-owner evidence.

Source: e1198952 / WORKPLAN.md / lines 51-75

RF-05 | HIGH | No independent scientific lift or transfer is established

Impact: All ten planned stage-2 intervals include zero on a development-exposed corpus. Neither replay nor new formatting creates external validation.

Fix / disposition: Retain full inference beside scientific proof cards; define DG-4 frozen new-panel/margin/replicate gate and unfavorable-result branch. [disclosed_future_proof]

Verification / remaining proof: No winner badge or ROI inference; all 12 cells and 10 contrasts remain linked unchanged.

Source: e1198952 / evidence_summary.json / lines 1-284

RF-06 | HIGH | No measured buyer value or willingness to pay

Impact: Workflow composition/provenance could be valuable, but there is no documented buyer interview, matched value study, adoption evidence or pricing evidence.

Fix / disposition: Commercial hypothesis ledger, current-alternative comparison and DG-3 decision protocol; unknown thresholds remain blocking and unestimated. [disclosed_future_proof]

Verification / remaining proof: No customer/pricing/savings invented; primary value metric and decision margin must be frozen before a pilot.

Source: e1198952 / WORKPLAN.md / lines 269-313

RF-07 | HIGH | License/provenance is not a chain-of-rights conclusion

Impact: Public visibility, existing proprietary terms and data attribution do not prove exclusive ownership or permission for every proposed reuse.

Fix / disposition: Contribution/provenance matrix, payload rights gate and candid unresolved ownership status; keep existing terms and original attribution. [disclosed_requires_owner]

Verification / remaining proof: Review each selected object, not a blanket claim that this packet resolves rights or patentability.

Source: e1198952 / DECISIONS.md / lines 1-23

RF-08 | MEDIUM | The moat is a hypothesis and adjacent foundations already exist

Impact: Workflow provenance and stateful agent orchestration are established adjacent capabilities. The packet does not prove competitor absence or exclusivity.

Fix / disposition: Primary-source AiiDA/LangGraph acknowledgement and contribution-specific matrix; measure claimed integration/value rather than marketing a universal first. [disclosed_future_proof]

Verification / remaining proof: No unsupported world-first/patent/moat assertion; source-backed present mechanism distinct from future commercial advantage.

Source: e1198952 / WORKPLAN.md / lines 184-205

RF-09 | MEDIUM | Checker subprocesses were weakly bounded and checks repeated

Impact: capture_output was memory-unbounded, parser accepted only the last output line, and verify-source ran capsule checks twice.

Fix / disposition: Bound stdout/stderr, reject malformed whole JSON, use a shared deadline and isolated child Python, and run the required capsule pass once. [resolved_in_scoped_local_tests]

Verification / remaining proof: Output flood, timeout, leading garbage JSON and duplicate execution regression cases fail safely.

Source: e1198952 / capsule.py / lines 27-39

RF-10 | MEDIUM | Task/JSON shape and pointer validation had gaps

Impact: Null arrays and wrong shapes could escape clean rejection; source ranges/phase membership and canonical array selectors were insufficiently checked.

Fix / disposition: Stricter collection/phase/evidence validation and canonical JSON-pointer array indexes; fixed safe CLI error output. [resolved_in_scoped_local_tests]

Verification / remaining proof: Null/malformed arrays, wrong phase membership, negative index and bad pointer escapes reject.

Source: e1198952 / core.py / lines 102-111

RF-11 | MEDIUM | Output aliases could evade the checkout exclusion

Impact: V1 used absolute paths and checked only the immediate output parent; an ancestor symlink could point back into the checkout.

Fix / disposition: Resolve output paths before checkout/inside-capsule checks and reject existing/linked destinations. Keep the trusted-directory/TOCTOU limit explicit. [resolved_in_scoped_local_tests]

Verification / remaining proof: Ancestor link into repository, existing output and archive-inside-payload aliases reject without overwriting.

Source: e1198952 / cli.py / lines 58-69

RF-12 | MEDIUM | Historical pass labels could be mistaken for current readiness

Impact: A fresh receipt check validates old assertions; it does not rerun old process/recovery scenarios or assess current installed service health.

Fix / disposition: Reports enumerate executed, historical-only and omitted checks; DG-2 requires current installed-owner proof separately. [resolved_in_scoped_local_tests]

Verification / remaining proof: Source and capsule scope fields differ correctly; no historical pass is labeled fresh process execution.

Source: e1198952 / capsule.py / lines 89-102

RF-13 | MEDIUM | Roles and effort ranges do not establish staffing or budget

Impact: A named role is not an assigned person. Uniform task-minute estimates and a six-hour aspiration do not establish delivery capacity.

Fix / disposition: Unassigned roles and unapproved budgets are explicit at every decision track; remove uncalibrated effort ranges and schedule promises. [implemented]

Verification / remaining proof: Activation of future work needs named ownership, a frozen scope and an approved budget; current task authorizes only bounded revision/publication.

Source: e1198952 / task_ledger.json / lines 1-80

RF-14 | MEDIUM | The additional reviewer did not inspect the delivered artifacts

Impact: The independent worker was denied Library artifact access and reviewed only supplied reconciliation. It cannot be called an independent full-PDF/code audit.

Fix / disposition: Separate review coverage statement; local preparer inspected actual artifacts and ran the confirmed code probes. [implemented]

Verification / remaining proof: Review credits state precisely which material was and was not accessible.

Source: e1198952 / HANDOFF.md / lines 1-32

Actual version 2 checks

49 focused tests: PASS. This qualifies the named tooling boundary; it is not a count of scientific or commercial risks retired.

Source/capsule report: verified. The original twelve historical assertions and PLGA aggregate check executed as stated in the receipt. Historical fitting/process behavior was not rerun.

Full commands, code hashes, environment, stdout digest, coverage and requested arithmetic outcome are in V2/evidence. The additional review worker did not inspect the inaccessible Library artifacts; local artifact and code review was performed by the preparer.

Decision track	State	New evidence required
DG-1 - Diligence inspectability	partial	Reviewed claim set; brief/systems note; bounded replay/integrity receipts; final freshness and rights scope.
DG-2 - Product end-to-end integration	unverified	Frozen tool packaging, Registry/Runtime admission, data rights, retained producer chain, installed recovery and exact output reconciliation for one scientific profile.
DG-3 - Workflow value against the current alternative	unverified	Named buyer/user/workflow and current alternative; predeclared matched tasks and value metric; measured effort, correctness and adoption constraints.
DG-4 - Independent future scientific outcome	not_met	Overlap-screened new-study panel, fixed sampling schedule and horizon, model/comparator, estimand, margin, missingness and assay-replicate uncertainty frozen before outcomes.
DG-5 - Defensibility, provenance and accountable release	partial	Contribution/foundation matrix; source and third-party provenance; actual rights/author approvals; named role assignments; authorized budget; narrow differentiation evidence.

The 93 task IDs remain in TASK_REGISTER.md and task_ledger.json. No task percentage is presented as investment readiness. Missing owners, budgets and decision margins are deliberately unresolved.

Source and artifact identity

The complete source lock and specification register accompany this document. Their hashes bind inspected inputs; they do not establish publisher authenticity or scientific validation.

SPEC-A | 1097 lines | 81314 bytes

Amphora_Investor_Technical_Diligence_Master_Engineering_Spec_2026-10-18.md

SHA-256: `847fab0c586fadec0b994dc954fec27dd82f3bae727d5fcac3796129f53f2bfc`

SPEC-B | 1035 lines | 84461 bytes

Amphora_Investor_Diligence_Master_Engineering_Spec_2026-10-18.md

SHA-256: `3e82eab88097ea926c9ee8780edd220d85229bc7da7965eaf1019204ffcc092a`

SPEC-C | 999 lines | 93841 bytes

Amphora_Investor_Technical_Diligence_Master_Spec_2026-10-18.md

SHA-256: `bc145f22eb834920b2878be996d47628d778d7f20e699c175c560496578f3592`

Repository basis: `9c3018dcdd17f40be27fc3c6dc7057d2d7cb05bc`. Original PLGA PDF: `f228757745d9d4836f6d048120867348ef6718e9763ca48ea0710f60b11dd33a`.

Fresh evidence reports and final branch/commit are in the handoff. An earlier review receipt cannot approve a changed PDF. No raw attachment, private correspondence or new scientific experiment is created by this workplan.